

Aman Behera

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Education

Indian Institute of Technology, Roorkee
Bachelor of Technology in Chemical Engineering

Oct 2022 - May 2026
CGPA: 8.23 / 10

Experience

Data Scientist (PPO) | HiLabs Inc

Feb 2026 - Present

- Developed **RDAN-GRPO**, a Rubric-based reinforcement learning method that improves instruction following by decoupling constraint compliance from response quality, improving Qwen3-4B by **15.42** on MulDimIF. **Hugging Face**
- Owned end-to-end development of **notSoSmartOCR**, combining orientation detection, layout checking, Falcon OCR 1.5, and selective LLM refinement to structure complex scans at **80% lower cost** than the legacy pipeline for document OCR.
- Architected and benchmarked a **Kubernetes-native llm-d/vLLM GPU serving platform** with precise KV-cache-aware routing, WVA/HPA autoscaling, Prometheus/Grafana observability, and **PostgreSQL-backed persistence**; reduced **p95 TTFT from 26.9s to 0.194s** and **end-to-end p95 latency from 31s to 0.79s** versus round-robin routing.
- Integrated **TypeSafe Jev** into a webpage-classification pipeline, improving accuracy from **88.52% to 96.67%** and macro F1 from **0.8823 to 0.9636** through refined label definitions and structured decision questions, compared to legacy XGB model

Data Science Intern | Info Edge India Ltd

May 2025 - Jul 2025

- Fine-tuned **Qwen3-8B-Instruct** on **27,000 GEO examples** using **SFT** and **DPO**, improving position-adjusted citation visibility by **37%** and **G-Eval LLM as a judge** citation quality from **0.43 to 0.77** across **600 held-out queries**.
- Designed a **position-aware citation metric** and seven-dimension **G-Eval** framework; evaluated **1500 generated responses** and improved citation correctness by **42 percentage points** while preserving fluency & factual consistency.

Projects

XpertGPT: Data-Constrained Pretraining with Multi-Scale Expert Routing

April 2026 - Aug 2026

- Designed and trained **XpertGPT**, a **52.6M-parameter sparse Transformer decoder** combining global self-attention, multi-scale sliding-window experts, and Expert-Choice routing on the **10M-word BabyLM Strict-Small corpus**
- Surpassed the **98.4M-parameter GPT-2** baseline on the aggregate BabyLM evaluation score (**38.42 vs. 37.38**) while using **46% fewer total parameters**, **67.3% fewer active parameters**, and **67.3% lower compute** per forward pass (**0.33 vs. 1.01 GFLOPs**), demonstrating a stronger performance-efficiency tradeoff under data-constrained pretraining.
- Achieved **balanced expert routing across all six layers** with zero token starvation and consistent token allocation; controlled ablations showed that placing global attention before expert routing is key for stable multi-task performance.

Research Publications

MycoGate-VL for Scorer-Aware Multilingual Hallucination Detection in VLMs

EMNLP 2026

Mixing Matters: Position Bias Across Sequence Mixers in Long-Context Answering

NeurIPS 2026

Post-Audit Mirage: Identifiability Limits of Offline Agent Verification

Under review, NeurIPS 2026

Academic Achievements & Competitions

- Secured **AIR 4759** in **JEE Advanced 2022** among 1.5 lakh candidates
- Won **Silver** at the **Inter IIT Tech Meet 2025**, leading a team of 6 from IIT Roorkee
- Placed **3rd** in the **Predictive AI Evaluation Competition** at **NeurIPS 2026**
- Selected for **Google Summer of Code 2026** as a contributor to **GNU Octave**
- Won the **HiLabs AIQuest Hackathon 2025** at IIT Roorkee and received Pre-placement and Internship offer

Olympiad Experience

- Awarded with the **NTSE Scholarship, 2020** by NCERT.
- Kishore Vaigyanik Protsahan Yojana, 2021**, **AIR 122** in the SA stream
- Indian Olympiad Qualifier in Chemistry (IOQC), 2021**, **top 300** nationwide
- Indian National Chemistry Olympiad (InChO), 2020**, **top 300** nationwide

Technical Skills

Python, C++, SQL, PyTorch, CUDA, SFT, RAG, Reinforcement Learning, LLM Evaluation, OCR, AI Agents, Agentic Memory

Positions of Responsibility

Project Head, ACM Student Chapter | IIT Roorkee

Apr 2025 - May 2026

- Led **50+ members** and organized talks, workshops, and projects in core computer science and machine learning.

Member, Vision and Learning Group | IIT Roorkee

Jan 2024 - May 2026

- Participated in research discussions, workshops, and projects focused on **deep learning, NLP, and computer vision**.

Competitive Programmer | Codeforces, LeetCode

Aug 2024 - Present

- Codeforces max rating **1743** (Expert), handle: **heWhoCooks**; LeetCode rating **1704**, handle: **beingamanforever**.